

Juan Miguel Badayos

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Profile

Software Engineering student with professional full-stack development experience (ASP.NET Core, C#, SQL Server, Vue.js, TypeScript) and testing experience across manual test design, test automation (Selenium, Robot Framework, pytest), API testing (Postman), performance testing (JMeter), security testing (OWASP ZAP, Bandit), and CI/CD test pipelines (GitHub Actions, Jira). Currently learning BDD and containerised test environments. Passionate about building reliable software and solving practical engineering problems.

Education

Charles Darwin University

March 2023 – November 2026 (expected)

Bachelor of Software Engineering Honours

- Awards & Scholarships: Outstanding Academic Achievement, NEC Information Technology Scholarship, Vocus Communications Scholarship, DXC Technology Scholarship, CDU Equity Award, NEC and Cisco Networking Scholarship, IT Code Fair 2024: Academic Excellence Award, IT Code Fair 2025: Academic Excellence Award
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Web Engineering, Distributed Systems, Software Architecture, Data Science

Technical Skills

Testing & QA: Manual test design (black-box, white-box, experience-based techniques), Selenium WebDriver, Robot Framework, pytest, Postman, Apache JMeter, OWASP ZAP, Bandit, GitHub Actions, Jira, Trello, defect tracking

Languages: C#, JavaScript/TypeScript, SQL, Python, Go, PHP, Bash

Framework & Tools: Django, .NET Core, Vue.js, NUnit, React Native, Node.js, WebSocket, Nginx, Git, Linux

Concepts: RESTful API design, concurrent programming, database design, authentication (OAuth), software testing, cloud deployment (Linux VPS)

QA / Software Testing — Academic Practice

SauceDemo Login Feature — Manual & Automated Test Suite (Python, Selenium, pytest, Postman, GitHub Actions, Jira)

- Wrote a test plan and 15 manual test cases (functional, negative, boundary) and logged defects in Trello.
- Automated 5 test cases with Selenium (Python/pytest) and wrote 5 Postman API tests checking status codes and response fields.
- Set up a GitHub Actions pipeline to run the Selenium tests automatically on every push.
- Migrated defects into Jira as tracked Bug issues with Priority assigned.

Performance & BDD Test Automation (Apache JMeter, Cucumber/Gherkin, Page Object Model)

- Built a JMeter load test with 50 concurrent users against a public REST endpoint, measuring throughput, error rate, and response time.
- Refactored the Selenium suite into a Page Object Model and wrote BDD scenarios in Gherkin/Cucumber for the login feature.

Mayan EDMS — Functional, Performance & Security Testing (Robot Framework, Selenium, JMeter, OWASP ZAP, Bandit)

- Owned testing for the document ingestion and classification features (upload, metadata, cabinets) of Mayan EDMS, an open-source document management system.
- Built an automated functional test suite in Robot Framework using the Requests, Selenium, and Database libraries; the suite caught a real API bug (duplicate cabinet creation not rejected), confirmed independently with curl.
- Ran JMeter load tests at 100 and 400 concurrent users against REST endpoints, analysing APDEX, response times, and error rates; identified a functional bug and a CPU-bound performance bottleneck.
- Performed security testing with OWASP ZAP (DAST) and Bandit (SAST), identifying vulnerabilities including a deprecated dependency and a potential SQL injection risk.

Experience

Radical Systems Pty Ltd

April 2026 – Present

Software Developer

- Developed backend functionality using ASP.NET Core, C#, and SQL Server.
- Built responsive frontend features using Vue.js and TypeScript.
- Wrote NUnit unit tests to improve backend reliability and prevent regressions.
- Diagnosed and resolved software bugs across existing applications.
- Implemented new features and enhancements for production applications based on development requirements.
- Collaborated with senior developers in an Agile environment, participating in sprint stand-ups and delivering assigned tasks.

NEC IT Solutions Australia Pty Ltd

November 2023 – April 2026

EUC Engineer

- Provided technical support for Windows environments, diagnosing OS, network, hardware, and application issues.
- Resolved user issues across Microsoft Office, desktop applications, and local network environments.
- Provided timely and effective status updates to customers.

KFC – Casuarina, NT

June 2022 – July 2024

Food Service Team Member

- Followed strict quality and safety standards and trained new staff.
- Prepared and cooked food according to quality standards.

Projects

Cozy Park – Multiplayer Real-Time Web Game (Go, WebSocket, Linux, Nginx)

- Engineered a server-authoritative Go WebSocket backend to coordinate real-time game state and minimise client desynchronisation.
- Designed a concurrency model to improve session stability and reduce perceived latency for simultaneous players.
- Deployed and maintained the services on a Linux VPS with Nginx reverse proxy.

Thesys – Thesis Management Web App (Django, Python, SQLite)

- Built a role-based access control system using Django to automate submission, review, and approval workflows for students and supervisors.

EchoLingo – Mobile App for Tiwi Language Learning (React Native, Firebase)

- Designed a React Native application integrated with Firebase for real-time teacher-student data synchronisation, preserving Tiwi language resources digitally.